

CALCULUS I, QUIZ-3, 02/04/26 Name: TEJAS SAH

- Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $f(x, y) = (x^2 - y^2, 2xy)$.
 - Show that f is locally invertible at every point in $\mathbb{R}^2 \setminus \{(0, 0)\}$.
 - Let $a = (1, 1), b = (0, 2)$. Find
 - the derivative of f at a , and
 - the derivative of g at b , where g is the local inverse of f near $(1, 1)$.
- Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $f(x, y) = (x + y^2, x^2 + y)$. Describe the locus of points in $\mathbb{R}^2$ where the derivative of f is not invertible.

Q1) Observe that f is a continuously differentiable over $\mathbb{R}^2$,

as $f'(x, y) = \begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix}$ is continuous over x, y as $D_1 f_1, D_2 f_1, D_1 f_2, D_2 f_2$ are continuous.

$\therefore f$ is continuously differentiable.

and observe that $\forall (x, y) \neq (0, 0)$, $|f'(x, y)| = (4x^2 + 4y^2) > 0$ strictly greater.

$\therefore f'(x, y)$ is non singular at all $(x, y) \neq (0, 0)$.

As $\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$, we can apply Inverse function theorem,

so for each $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$, we have an open ball around it $U_{x, y}$, whose image in $\mathbb{R}^2 = V_{x, y}$ is open.

and $f|_{U_{x, y}}: U_{x, y} \rightarrow V_{x, y}$ is a diffeomorphism $\rightarrow$ hence an inverse exists for $f|_{U_{x, y}}$, and so

f is locally invertible at every point in $\mathbb{R}^2 \setminus \{(0, 0)\}$.

Rough

$$f'(x, y) = \begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = (ad-bc)I$$

b) $a = (1, 1), b = (0, 2)$

i) Find $f'(a) = f'(1, 1) = \begin{bmatrix} 2 & -2 \\ 2 & 2 \end{bmatrix}$

(Also $f(1, 1) = (0, 2)$)

ii) Find $g'(b)$, where g is local inverse of f near $(1, 1)$

$\rightarrow$ find a small neighbourhood around a : U_a , the map $f|_{U_a}: U_a \rightarrow V_a$ is diffeomorphic, and its inverse

g has the derivative: $g'(x, y) = [f'(g(x, y))]^{-1}$ (by IFT)

As $f(1, 1) = (0, 2) \rightarrow g(0, 2) = (1, 1)$ (local inverse)

$$\therefore g'(0, 2) = [f'(1, 1)]^{-1} = \begin{bmatrix} 2 & -2 \\ 2 & 2 \end{bmatrix}^{-1} = \frac{1}{8} \begin{bmatrix} 2 & 2 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} \end{bmatrix}$$